LiDAR Maze 10/21

Software + FPGA

Project overview

- This is the second quarter of this project
- We have some code + designs from last quarter
- More hands-on engagement this quarter
- We are building a small “car” (~10x10 inches) to autonomously solve a maze
- We must build the car and the maze
- Will be using github

Raise of hands: new to this project?



The roof-mounted unit installed on Waymo's ride-hailing vehicles contains various hardware components for autonomous driving. *Courtesy of Waymo*

hi

- Pair up with one person
- One person holds the pen, they must close their eyes
- The other has to instruct them through the maze
- 2:30 (?) timer

Leads reminder: take pictures

Part 2

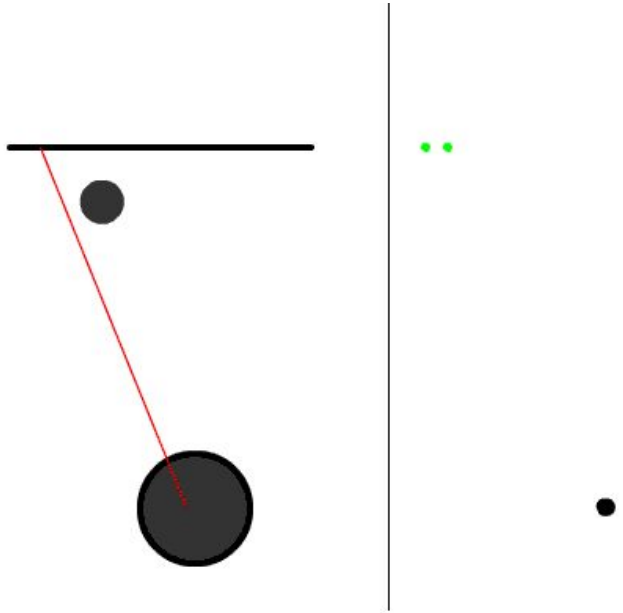
- Find two other new people !
- One of you is blind and holds the pen
- One of you is blind but cannot look at the maze (only the third person)
- One person looks at the maze, but they cannot speak (use hand signs)
- 2:30 (?) timer

Leads reminder: take pictures

Software subteam

- Write maze-solving algorithms and test them, digitally and physically.
- Create a digital simulation of the car and maze environment to test different solving algorithms

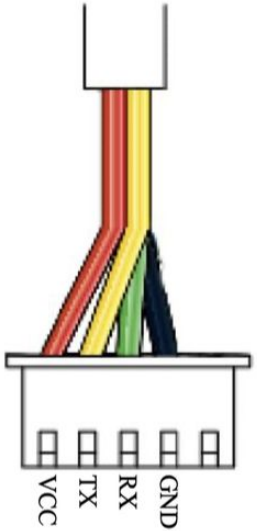
Leads: Akshay

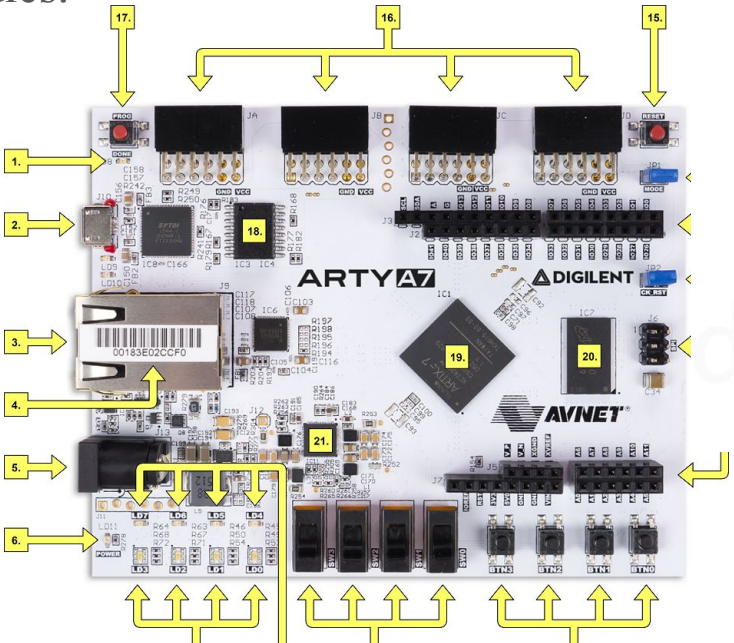


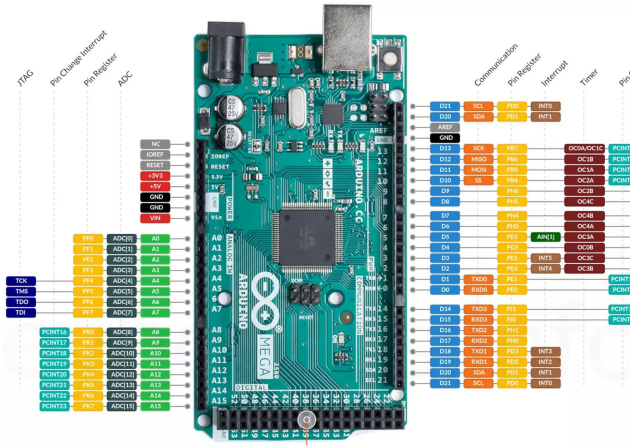
FPGA subteam

Leads: Alexa, Dara

- Implement LiDAR -> FPGA -> Arduino data pipeline (including wiring)
- Write SystemVerilog code: LiDAR initialization, data packaging, and data transmission modules.







Meeting today

- New members must complete a hands-on task to join their desired subteam
- Use the internet, AI, whatever you want to complete it
- PLEASE ask the leads questions (this isn’t a test! It shouldnt be silent in this room)
- This is to emulate how meetings will go. Leads are here to give you a task and help you, not be your tutor.
- You will have until the end of the meeting

task

Software: In groups: research and come up with a maze-solving algorithm that we could use. Imagine you are the robot solving a maze with the given data using your algorithm.

FPGA: Look up RP LiDAR C1, Arty A7 FPGA, Arduino Mega 2600 communication pinouts (datasheets). What do they mean? Determine how to wire them together.

Any questions? Ask your lead where to start!

(demo the lidar)

- Rn the lidar is connected to computer
- Eventual goal is lidar -> fpga -> arduino
- The fpga acts as a data repackager/processor
  - (enables parallel pathfinding and navigation on arduino)
  - Industry standard
  - yes
- Any questions?

task

Software: In groups of 3+: research and come up with a maze-solving algorithm that we could use. Imagine you are the robot solving a maze with your algorithm.

FPGA: Look up RP LiDAR C1, Arty A7 FPGA, Arduino Mega 2600 communication pinouts (datasheets). Determine how to wire them together.

Any questions? Ask your lead where to start!

meeting wrap up

- Raise of hands: who would be interested in a soldering workshop?
- Any other questions? leads will stick around (introduce yourself)

attendance

Miss one: warning

Miss two: speak to lead

Miss three: kicked from project

Please communicate your absences! We are flexible

Code: wart

